

lab2double Function Test Results

Test Case 1: Typical uint8 LAB Image

"Input type:" "uint8"

"Input:" uint8(cat(3, ...
[0 128; 255 64], ...
[128 148; 108 168], ...
[128 98; 158 188]))

"Output type:" "constant"

"Output:"

(:,:,1)

0. 50.196078

100. 25.098039

(:,:,2)

0. 20.

-20. 40.

(:,:,3)

0. -30.

30. 60.

Test Case 2: Minimum uint8 Values

"Input type:" "uint8"

"Input:" uint8(zeros(2,2,3))

"Output type:" "constant"

"Output:"

(:,:,1)

0 0
0 0
(:,:,2)
-128. -128.
-128. -128.
(:,:,3)
-128. -128.
-128. -128.

Test Case 3: Maximum uint8 Values

"Input type:" "uint8"

"Input:" uint8(255 * ones(2,2,3))

"Output type:" "constant"

"Output:"

(:,:,1)
100. 100.
100. 100.
(:,:,2)
127. 127.
127. 127.
(:,:,3)
127. 127.
127. 127.

Test Case 4: Typical uint16 LAB Image

"Input type:" "uint16"

```
"Input:" uint16(cat(3, ...  
    [0 32640; 65280 16320], ...  
    [32640 40000; 20000 50000], ...  
    [32640 30000; 45000 60000]]));
```

"Output type:" "constant"

"Output:"

(:,:,1)

0. 50.

100. 25.

(:,:,2)

-0.5 28.25

-49.875 67.3125

(:,:,3)

-0.5 -10.8125

47.78125 106.375

Test Case 5: Minimum uint16 Values

"Input type:" "uint16"

"Input:" uint16(zeros(2,2,3))

"Output type:" "constant"

"Output:"

(:,:,1)

0. 0.

0. 0.

(:,:,2)

-128. -128.

-128. -128.

(:,:3)

-128. -128.

-128. -128.

Test Case 6: Maximum uint16 Values

"Input type:" "uint16"

"Input:" uint16(65280 * ones(2,2,3))

"Output type:" "constant"

"Output:"

(:,:1)

100. 100.

100. 100.

(:,:2)

127. 127.

127. 127.

(:,:3)

127. 127.

127. 127.

Test Case 7: Double Input

"Input type:" "constant"

"Input:"

cat(3, ...

[50 75; 25 100], ...

[0 20; -20 40], ...

[0 -30; 30 60]);

"Output type:" "constant"

"Output:"

(:,:1)

50. 75.

25. 100.

(:,:2)

0. 20.

-20. 40.

(:,:3)

0. -30.

30. 60.

Test Case 8: Single Pixel

"Input type:" "constant"

"Input:" cat(3, 50, 0, 0)

"Output type:" "constant"

"Output:"

(:,:1)

50.

(:,:2)

0.

(:,:3)

0.

Test Case 9: Mx3 Double Colormap

"Input type:" "constant"

"Input:" lab = [...

0 -128 -128;

50 0 0;

100 127 127];

"Output type:" "constant"

"Output:"

0. -128. -128.

50. 0. 0.

100. 127. 127.

Test Case 10: Invalid 2-D Input

"Input type:" "uint8"

"Input:" uint8([1 2; 3 4])

"Output:" "lab2double: LAB must be Mx3, MxNx3, or MxNx3xK size"

Test Case 11: int16 Input

"Input type:" "int16"

"Input:" int16(cat(3, ...

[0 50], ...

[0 0], ...

[0 0]));

"Output:" " lab2double: invalid class int16 for LAB"